

# Representing inputs as vectors

Project: Image-pattern Network Investigation

## 1. Prepare the purpose, prerequisites and materials

### Teach the learning cycle

Primary target: Encode a record using a fixed feature order and units

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

---

**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

---

**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

---

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

## Retrieval answer guidance

### ML Builder Expo

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

Look for: Defend the project using traceable evidence

Reference example, not the only acceptable wording: Teacher explanation: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

Begin Grade 9 by connecting the Grade 8 feature table to an ordered numerical input. Students should name every component before calculating. Use two 45-minute sessions. Avoid treating a vector as a mysterious neural object: the learning target is a precise data contract, including order, units, missing values and the distinction between inputs and targets.

### Outcomes

Encode a record using a fixed feature order and units

Distinguish a vector dimension from the number of examples

Detect missing values and target leakage before computation

### Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

### Prerequisite answer

The class is the target being predicted. Supplying that answer as an input leaks information that would not be available for a new prediction.

### End goal for the pictorial case

By the end, I can explain representing inputs as vectors using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain and model the mechanism

A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields. A vector is an ordered list: [4, 2] can mean

width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same schema.

For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The vector has dimension 2 because it contains two components. Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

There are three examples, each with two features. Stacking their vectors makes a table with three rows and two columns. Do not confuse the number of rows with the dimension of one input.

Read fields by name, convert to the declared units, validate, and only then write the vector. Parcel D has width 50 mm and height 30 mm. Because 10 mm equals 1 cm, its encoded vector is [5, 3]. Writing [50, 30] would preserve the numbers while changing the physical meaning by a factor of ten.

This short function assumes each supplied value is a number in mm or None for missing. It is a classroom example, not a complete input validator. Trace the conversion before running it. The function represents data; it does not fit a model or predict a class.

### Term · Meaning

**Term:** Vector

**Meaning:** An ordered list of numerical components

---

**Term:** Component

**Meaning:** One value at a specified position in a vector

---

**Term:** Dimension

**Meaning:** The number of components in a vector

---

**Term:** Schema

**Meaning:** The recorded feature order, units and validity rules

---

## An input vector keeps an agreed order

### Source record · Encoding action · Model input

**Source record:** D width 50 mm

**Encoding action:** Divide width by 10

**Model input:** First component 5

---

**Source record:** D height 30 mm

**Encoding action:** Divide height by 10

**Model input:** Second component 3

---

**Source record:** Schema width then height

**Encoding action:** Keep named positions

**Model input:** Vector [5, 3] in cm

---

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Guided E: [2, 7]. The reversed [7, 2] means width 7 cm and height 2 cm, a different measurement pair.

Exit: [4, 2] reveals only width and height under the schema. It cannot reveal the omitted distinction causing different labels; the same deterministic input receives the same output.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

## An input vector keeps an agreed order

**Source record · Encoding action · Model input**

**Source record:** D width 50 mm

**Encoding action:** Divide width by 10

**Model input:** First component 5

---

**Source record:** D height 30 mm

**Encoding action:** Divide height by 10

**Model input:** Second component 3

---

**Source record:** Schema width then height

**Encoding action:** Keep named positions

**Model input:** Vector [5, 3] in cm

---

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

**Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.**

---

---

---

---

**Thinking move · Teacher prompt**

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

---

**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

---

**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

---

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## **A pictorial worked case: Representing inputs as vectors**

Encode the same pixel measurements in a fixed vector order.

### **The facts we will use**

Toy image features are [left\_brightness, right\_brightness], each = 0..10. ImageA has left2, right7; imageB has left7, right2.

AI ACADEMY / AI-5/1

## Representing inputs as vectors

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Declare coordinate order

Coordinate0 is left brightness; coordinate1 is right brightness.



#### 2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.



#### 3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

### CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Declare coordinate order

**What happens and why:** Coordinate0 is left brightness; coordinate1 is right brightness.

---

**Step:** 2. Encode both images

**What happens and why:** A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns

differ.

---

**Step:** 3. Check input contract

**What happens and why:** Use the same order when training and predicting; record missing measurements rather than guessing.

---

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

A vector is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter. A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

### What this example does not establish

These two features are simplified measurements, not a complete image representation or evidence of learned vision.

### Change one thing in the pictured case

Change: Swap only A's vector order before prediction.

Prediction: The encoded input becomes [7, 2], numerically identical to B's original input.

Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

#### Time or stage · Teacher action

**Time or stage:** Session 1 start to 7 minutes

**Teacher action:** Ask the prerequisite question. Retrieve the distinction between a feature and a target; require a prediction-time explanation.

---

**Time or stage:** 7 to 18 minutes

**Teacher action:** Place A, B and C on paper. Students label horizontal width and vertical height, then write vectors with the schema above them.

---

**Time or stage:** 18 to 32 minutes

**Teacher action:** Complete W1 and W2. Contrast three examples with two components per example. Model the mm-to-cm conversion explicitly.

---

**Time or stage:** 32 to 45 minutes

**Teacher action:** Trace the Python function or its paper equivalent. Complete the guided E case and compare the swapped vector.

---

**Time or stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the schema without looking. Ask how two correctly sized vectors can have different meanings.

---

**Time or stage:** 8 to 20 minutes

**Teacher action:** Use W3 and the identical-input example. Discuss valid zero in another schema without changing the parcel rule.

---

**Time or stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before giving hints. Check field order, dimensions, valid age zero and the excluded future target.

---

**Time or stage:** 35 to 45 minutes

**Teacher action:** Use W5 and the exit response. Give targeted feedback, then administer the fresh retry and retain both attempts.

---

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide empty boxes labelled with feature names and units. Accept a correct spoken explanation before written notation. Extension: propose two different real objects that share the same two-feature vector, then identify an additional measurable feature available before prediction.

## Reduce help deliberately

Completion task: A new record has width 60 mm and height 10 mm. Give its vector in the agreed cm schema, and name one error in [60, 10].

Teacher check: Teacher explanation: The vector is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Representing inputs as vectors. Proposed explanation: "Order does not matter". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A and B use the same numbers but assign them to different features. Read each position by name.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial practice question

Why does equal total brightness not make A and B identical inputs? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

## Reasoned teacher answer

A and B place brightness in different positions. The ordered vectors distinguish left/right structure; their common sum 9 loses that distinction. Changed-case check: The encoded input becomes [7, 2], numerically identical to B's original input. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Changed-case reasoning: The encoded input becomes [7, 2], numerically identical to B's original input. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

# 4. Run the investigation and diagnose misconceptions

Order does not matter: A and B use the same numbers but assign them to different features. Read each position by name.

More rows increase vector dimension: Rows count examples; components count features of one example.

A missing value is zero: Whether zero is valid depends on the field. Missing means unknown, and is not interchangeable with a measured zero.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

## Match the support to the first error

### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

---

**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

---

**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

---

**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

---

**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

---

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Guided and exit questions

Guided E: [2, 7]. The reversed [7, 2] means width 7 cm and height 2 cm, a different measurement pair.

Exit: [4, 2] reveals only width and height under the schema. It cannot reveal the omitted distinction causing different labels; the same deterministic input receives the same output.

## W1 Encode the records

Write A, B and C as ordered pairs from the lesson. State the dimension of each and the shape of the stacked table.

Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

## W2 Convert before encoding

A new record has width 60 mm and height 10 mm. Give its vector in the agreed cm schema, and name one error in [60, 10].

Teacher explanation: The vector is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

## W3 Handle missing information

Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this schema.

Teacher explanation: Missing is not measured zero. Height must be present and positive; reject the record and request a valid measurement before encoding.

## W4 Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

Teacher explanation: K [18, 5, 12], L [20, 4, 0]. Dimension 3, table 2 rows by 3 columns. Age zero is a valid known value. Exclude the future failure target from inputs; it is not available at prediction time.

## W5 Explain a collision

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two inputs apart.

Teacher explanation: The two input vectors are identical, so the fixed deterministic computation returns the same output. Additional rows do not create a missing distinction in these two vectors; a relevant available feature or an explicit uncertainty policy may be needed.

## Independent assessment and fresh retry

Assess W4 for named order, all values retained, dimension 3, two-row shape and exclusion of the future label. Fresh retry: schema [mass in g, length in cm]; two records (length 4, mass 12) and (mass 9, length 6) encode as [12, 4] and [9, 6], dimension 2, table 2 by 2. Ask why an absent mass must not become zero without a declared policy.

## Assess independent transfer

Selected case: A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

## Selected case answer

Teacher explanation: K [18, 5, 12], L [20, 4, 0]. Dimension 3, table 2 rows by 3 columns. Age zero is a valid known value. Exclude the future failure target from inputs; it is not available at prediction time.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

---

**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

---

**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

---

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect the project, support and next lesson

Add a labelled representing inputs as vectors evidence sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide empty boxes labelled with feature names and units. Accept a correct spoken explanation before written notation. Extension: propose two different real objects that share the same two-feature vector, then identify an additional measurable feature available before prediction.

Week 2 uses the ordered components in dot products. Revisit order and signed multiplication before introducing weights.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-5/2, ai-5/4, ai-5/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

#### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 7. Prepare, teach and assess

Teaching target: Encode a record using a fixed feature order and units

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

\_\_\_\_\_

**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

\_\_\_\_\_

**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38:

connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

---

Before class, solve the supported practice: A new record has width 60 mm and height 10 mm. Give its vector in the agreed cm schema, and name one error in [60, 10].

Reasoned practice answer: Teacher explanation: The vector is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

Check every learner before discussion: Use the worked case for Representing inputs as vectors. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A and B use the same numbers but assign them to different features. Read each position by name.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Swap only A’s vector order before prediction. Require a prediction and a reason before comparing. Expected result: The encoded input becomes [7, 2], numerically identical to B’s original input. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

### **Criterion · Not yet · With support · Independent**

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

---

**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

---

**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

---

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

## 8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.